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Requirements Gathering

جمع المتطلبات

GlobeClass — AI-Powered Satellite Image Land Cover Classification

1. Functional Requirements

Functional requirements define the specific behaviors and features the system must provide.

1.1 Image Classification Engine

ID	Requirement	Priority	Description
FR-01	Image Upload	High	The system shall allow users to upload satellite images in JPG and PNG formats for classification.
FR-02	Multi-Model Inference	High	The system shall support classification using three distinct neural network architectures: ResNet50, Swin Transformer, and Prithvi-EO 2.0.
FR-03	Model Switching	High	The system shall allow users to switch between available models in real-time via a dropdown selector.
FR-04	Classification Output	High	The system shall return the predicted land cover class, confidence score (%), and probability distribution across all 10 classes for every classification request.

FR-05	Image Preprocessing	High	The system shall automatically resize uploaded images to the required input dimensions (64x64 for ResNet50, 224x224 for Swin and Prithvi) and normalize pixel values.
FR-06	Drag-and-Drop Upload	Medium	The system shall support drag-and-drop file upload in addition to the standard file browser dialog.
FR-07	Image Preview	Medium	The system shall display a preview of the uploaded image with filename and file size before classification.

1.2 Live Satellite Capture

ID	Requirement	Priority	Description
FR-08	Interactive 3D Globe	High	The system shall render an interactive 3D Earth globe using Three.js with rotation, zoom, and coordinate tracking.
FR-09	Satellite Map View	High	The system shall provide a satellite map view using Leaflet.js with ESRI World Imagery tiles for high-resolution navigation.
FR-10	Globe-to-Satellite Transition	Medium	The system shall provide a smooth animated transition between the 3D globe view and the satellite map view.
FR-11	Satellite Image Capture	High	The system shall capture a 512x512 pixel satellite image from the current map viewport using the ESRI ArcGIS REST API.
FR-12	Capture & Classify	High	The system shall automatically classify a captured satellite image immediately after capture, displaying results in real-time.
FR-13	Coordinate Input	Medium	The system shall allow users to manually enter latitude and longitude coordinates to navigate to a specific location.
FR-14	Location Search	Medium	The system shall support searching for places by name (city, landmark, address) using the Nominatim geocoding API.

FR-15	GPS Geolocation	Medium	The system shall support using the browser's geolocation API to navigate to the user's current location.

1.3 Results & Visualization

ID	Requirement	Priority	Description
FR-16	Primary Result Display	High	The system shall display the top predicted class with its icon, label, description, and confidence percentage.
FR-17	Full Breakdown	High	The system shall show a ranked list of all 10 classes with individual confidence bars and percentages.
FR-18	Confidence Chart	Medium	The system shall render an interactive bar chart (Chart.js) visualizing the confidence distribution across all classes.
FR-19	Classification History	Medium	The system shall maintain a session-based classification log showing previous results with timestamps.
FR-20	Clear History	Low	The system shall allow users to clear the classification history log.

1.4 User Interface & Experience

ID	Requirement	Priority	Description
FR-21	Landing Page	High	The system shall have a landing page showcasing the project overview, model architectures, workflow, and land cover classes.
FR-22	Model Carousel	Medium	The landing page shall feature an interactive carousel displaying details for all three neural network architectures.
FR-23	Bilingual Support	Medium	The system shall support English and Arabic language switching with persistent language preference (localStorage).

FR-24	PWA Installation	Low	The system shall be installable as a Progressive Web App on supported browsers and devices.
FR-25	Sample Gallery	Medium	The system shall provide a gallery of sample test images from the EuroSAT dataset for quick testing.
FR-26	Model Info API	Low	The system shall expose an API endpoint (/api/model-info) returning metadata about all available models.

2. Non-Functional Requirements

Non-functional requirements define quality attributes and constraints the system must satisfy.

ID	Category	Requirement	Target
NFR-01	Performance	Classification inference latency	< 50ms for ResNet50, < 100ms for Swin/Prithvi (CPU)
NFR-02	Performance	API response time (end-to-end)	< 2 seconds including image preprocessing
NFR-03	Performance	Frontend page load time	< 3 seconds on broadband connection
NFR-04	Accuracy	ResNet50 test set accuracy	≥ 95% (Achieved: 98.25%)
NFR-05	Accuracy	Swin Transformer test set accuracy	≥ 95% (Achieved: 98.42%)
NFR-06	Accuracy	Prithvi-EO test set accuracy	≥ 85% (Achieved: 91.23%)
NFR-07	Scalability	Concurrent API requests	Support single-user inference with model hot-swapping
NFR-08	Usability	Mobile responsiveness	Fully functional on devices ≥ 320px width

NFR-09	Usability	Accessibility	Semantic HTML, proper heading hierarchy, descriptive alt texts
NFR-10	Security	Input validation	Validate image MIME types, reject non-image uploads
NFR-11	Security	CORS policy	Flask-CORS configured for authorized origins
NFR-12	Reliability	Health check endpoint	/health endpoint returns status for monitoring
NFR-13	Maintainability	Code organization	Modular architecture: separate frontend, backend, notebooks, models
NFR-14	Portability	Cross-browser compatibility	Chrome, Firefox, Safari, Edge (latest versions)
NFR-15	Availability	Service uptime	Railway deployment with automatic restarts
NFR-16	Memory Efficiency	Model loading strategy	Load only one model at a time; GC previous model to prevent OOM

3. User Requirements

3.1 User Personas

Persona 1: Environmental Researcher

- Wants to quickly classify satellite imagery for land cover mapping studies
- Needs high accuracy and detailed confidence breakdowns
- Values the ability to compare multiple model architectures

Persona 2: Student / Learner

- Wants to understand how deep learning works for satellite imagery
- Appreciates the interactive 3D globe and step-by-step workflow explanation
- Uses the sample gallery for quick experimentation

Persona 3: General User / Curious Explorer

- Wants to explore Earth and classify their local area
- Uses the GPS "My Location" feature and place search
- Appreciates the bilingual (EN/AR) interface and mobile PWA

3.2 User Stories

ID	As a...	I want to...	So that...
US-01	User	Upload a satellite image and get its land cover classification	I can quickly identify terrain types
US-02	User	Zoom into the 3D globe and transition to a satellite view	I can visually explore any location on Earth
US-03	User	Capture a live satellite image from the map and classify it	I can classify any real-world location
US-04	User	Switch between different AI models	I can compare classification results
US-05	User	See confidence scores for all 10 classes	I can understand how certain the AI is
US-06	User	Search for a city or landmark by name	I can quickly navigate to specific locations
US-07	User	Use my GPS location	I can classify the land cover at my current position
US-08	User	View sample images from the dataset	I can test the system without my own images
US-09	User	Switch the interface language to Arabic	I can use the application in my preferred language
US-10	User	Install the app on my phone	I can access it quickly like a native app
US-11	User	View my classification history	I can compare past results

4. Use Case Diagram

The diagram below illustrates all major use cases for the GlobeClass system and the interactions available to the User actor.

Use Cases:

- Upload Satellite Image
- Classify Image (Select Model: ResNet50 / Swin / Prithvi)
- View Results & Breakdown
- Explore 3D Globe
- Open Satellite Map View
- Capture & Classify Live Image
- Search Location by Name
- Use GPS / My Location
- Switch Language (EN / AR)
- View Classification History
- Install as PWA

Use Case Details

UC-01: Classify Uploaded Image

Actor: User

Precondition: User has a satellite image file (JPG/PNG)

Main Flow:

1. User opens the Observatory page
2. User drags an image onto the upload zone (or clicks to browse)
3. System displays image preview with filename and size
4. User selects a model from the dropdown (optional, defaults to ResNet50)
5. User clicks "Classify"
6. System sends image to the backend API
7. System displays the predicted class, confidence score, full breakdown, and chart

Postcondition: Classification result displayed; entry added to history log

UC-02: Capture & Classify Live Satellite Image

Actor: User

Precondition: User is on the Observatory page

Main Flow:

8. User interacts with the 3D globe (drag to rotate, scroll to zoom)
9. User clicks the globe or "Open Live Satellite View" button
10. System transitions to satellite map view (Leaflet + ESRI tiles)
11. User navigates to desired location
12. User clicks "Capture & Classify"
13. System fetches a 512x512 satellite image from ESRI ArcGIS
14. System automatically classifies the captured image
15. System displays results with coordinates

Postcondition: Captured image displayed; classification result shown

5. Data Sources

5.1 Training Data

Dataset	Source	Details
EuroSAT RGB	Sentinel-2 satellite (ESA)	27,000 labeled satellite images across 10 land cover classes. Each image is 64x64 pixels in RGB format. Data was collected from the Sentinel-2 multi-spectral imaging mission covering European territories.

Data Split:

Split	Purpose	Classes
Train/	Model training	10 folders (one per class)
Valid/	Validation during training (hyperparameter tuning)	10 folders (one per class)

Test/	Final model evaluation (unseen data)	10 folders (one per class)

The 10 Land Cover Classes:

#	Class Name	Description
1	AnnualCrop	Seasonal farming fields with crop rotation patterns
2	Forest	Dense tree canopy with high vegetation index
3	HerbaceousVegetation	Wild grassland or bushy terrain coverage
4	Highway	Paved road infrastructure and transport corridors
5	Industrial	Factories, warehouses, and commercial structures
6	Pasture	Open grazing land and meadow terrain
7	PermanentCrop	Vineyards, orchards, and long-term plantations
8	Residential	Urban housing and neighborhood developments
9	River	Natural flowing water bodies and waterways
10	SeaLake	Oceans, lakes, and large stationary water bodies

5.2 Real-Time Data

Source	API	Purpose
ESRI ArcGIS World Imagery	REST API (MapServer/export)	Fetching live satellite images for any GPS coordinates at configurable zoom levels (10-18)
Nominatim (OpenStreetMap)	REST API (search endpoint)	Geocoding — converting place names to latitude/longitude coordinates

Browser Geolocation API	HTML5 Geolocation	Obtaining the user's current GPS position for location-based classification

5.3 Pre-trained Model Weights

Source	Model	Purpose
ImageNet (ILSVRC)	ResNet50 pre-trained weights	Transfer learning backbone initialization
ImageNet-22K	Swin-Base pre-trained weights	Transfer learning for the Swin Transformer
NASA HLS (via terratorch)	Prithvi-EO 2.0 weights	Foundation model pre-trained on global satellite imagery